Presented by:
Aashma Varma

Beyond Price Tags

Do price and ratings reflect ingredient concern in skincare products?

Consumers often rely on price because ingredient lists are hard to interpret.

Ingredient lists are available but difficult to understand.

Price is the most accessible quality signal at the time of decision.

This raises the question of whether higher-priced products offer better formulation quality in reality.

Making the Right Choice

For the average customer, formulation quality is a criteria which they cannot evaluate on the spot. Hence, they rely on external cues such as price, branding & marketing.

Studies on consumer behaviour suggest that people often associate higher prices with higher quality.

This price-quality perception is especially relevant for non-durable goods like skincare, where product quality is difficult to evaluate before purchase.

Consumer autonomy does not end with having information alone; the information given must be accessible and understandable to make an informed purchase decision.

Research Questions

Do higher-priced products have lower ingredient concern scores?

Do higher-priced products receive higher ratings regardless of concern score?

Hypothesis:
Premium products show lower concern scores and maintain high ratings.

Product Dataset

Data was collected in two phases. The first dataset contained product-level information from Sephora skincare listings.

About the dataset

- Kaggle dataset of 1,400+ skincare products sold through Sephora.
- Included variable such as price, rating, product category and list of ingredients.
- Dataset reflects product information from 2023.

Limitations of the dataset

- Products are limited to those sold through online Sephora retailers, does not represent the skincare market entirely.
- Region-specific to the USA and missing many brands popular in larger skincare markets, especially East Asia.
- Ingredient data was incomplete and required manual review.
  - 134 products had missing ingredient lists
  - 77 were manually reviewed and filled
  - 10-12 products were excluded because information was unavailable
- Dataset only represents a snapshot in time. Ratings, price and formulation can change over time.

Flagged Ingredient Dataset

How it was built

- Manually selected ingredients commonly associated with irritation, sensitivity, debated safety, or toxicological concern
- Used public ingredient information from sources such as CIR reports and INCIdecoder
- Each ingredient was assigned a severity score based on the type and strength of concern

Score	Meaning	Rationale
1	Low concern	Ingredients mainly associated with mild irritation, dryness, sensitivity, or consumer preference concerns
2	Moderate concern	Ingredients with more debated safety concerns, stronger irritation potential, or controversy in cosmetic use
3	High concern	Ingredients with stronger toxicological concern, regulatory restrictions, carcinogenic contamination concerns, or ingredients generally avoided due to higher safety risks

The presence of a flagged ingredient does not automatically mean a product is unsafe, because risk also depends on concentration, formulation, skin type, and usage context.

Scoring Method

For each product, an ingredient concern score was calculated.

$$\frac{Total\ Ingredients - Position\ of\ Ingredient}{Total\ no.\ of\ Ingredients} \times Severity\ score\ of\ Ingredient$$

Example:

Ingredient List:
Aqua, Parfum, Mineral Oil $\longrightarrow$ Parfum is flagged with a severity score of 2, at position 2 out of 3 ingredients. $\longrightarrow \frac{3-2}{3} \times 2 = \frac{2}{3} = 0.67$

Since the ingredient list is written in decreasing order of concentration, flagged ingredients appearing earlier in the list receive a higher weighted score.

Strengths & Limitations

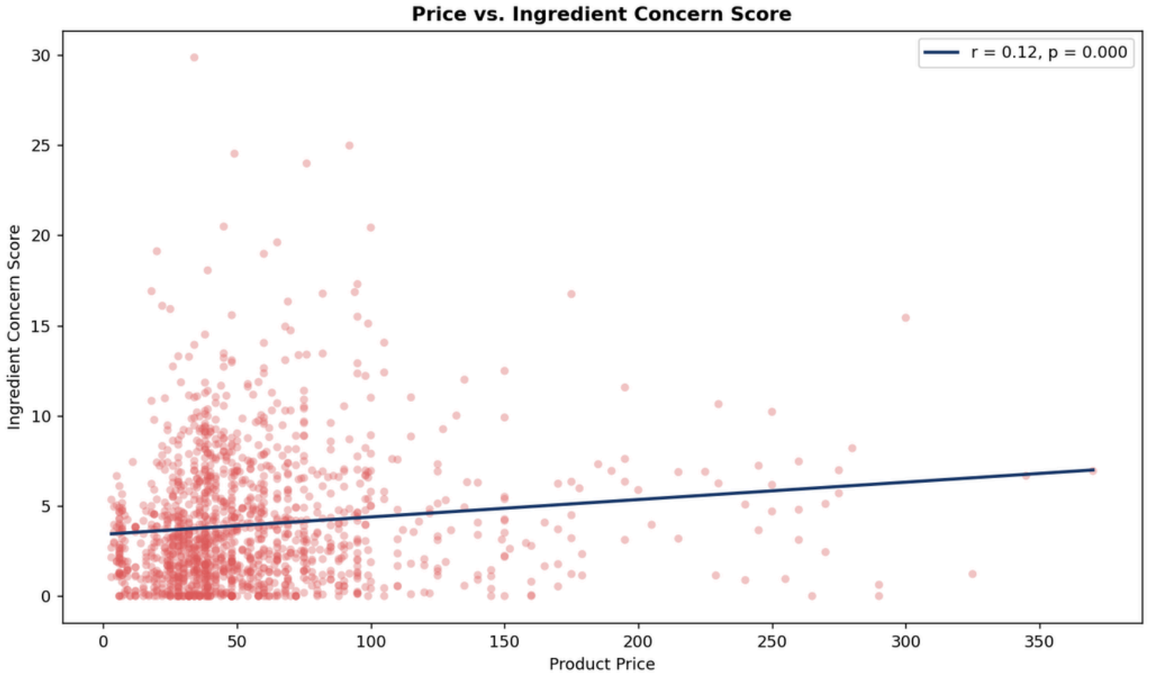
Strengths

- Simple and interpretable method for comparing products across the dataset.
- Position-weighted, so flagged ingredients appearing earlier in the ingredient list contribute more to the score.
- Provides a proxy for formulation concern, based on whether products use more controversial or potentially irritating ingredients.

Limitations

- The score does not capture full formulation context, including how ingredients interact with each other.
- Some flagged ingredients may be chosen because they improve stability, texture, preservation, or product performance.
- A higher score does not automatically mean a product is harmful, especially since safety depends on concentration, usage, and skin type.

Finding 1



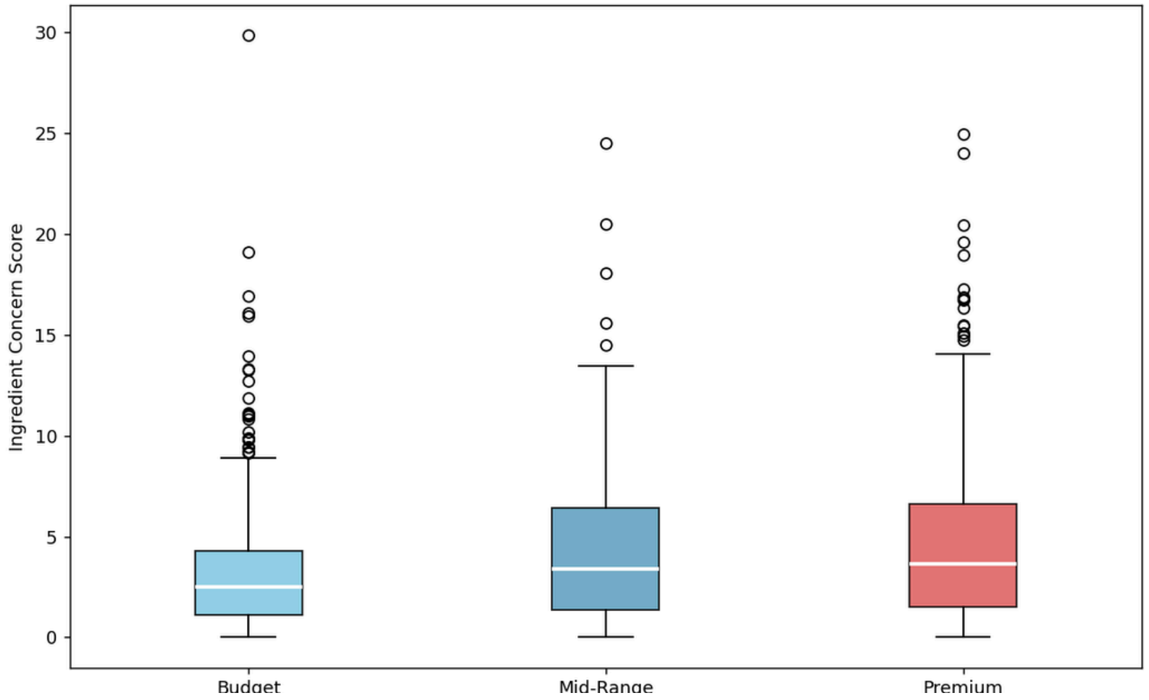
Very weak positive relationship (r=0.12) between price and concern score.

Statistically significant ($p<0.001$), but price explains only about 1.4% of variation ($R^2=0.014$).

Most products are clustered below $100, with fewer high-end products.

Implication: Higher price does not clearly indicate better or worse formulation. Price alone tells consumers very little about ingredient concern.

Finding 1



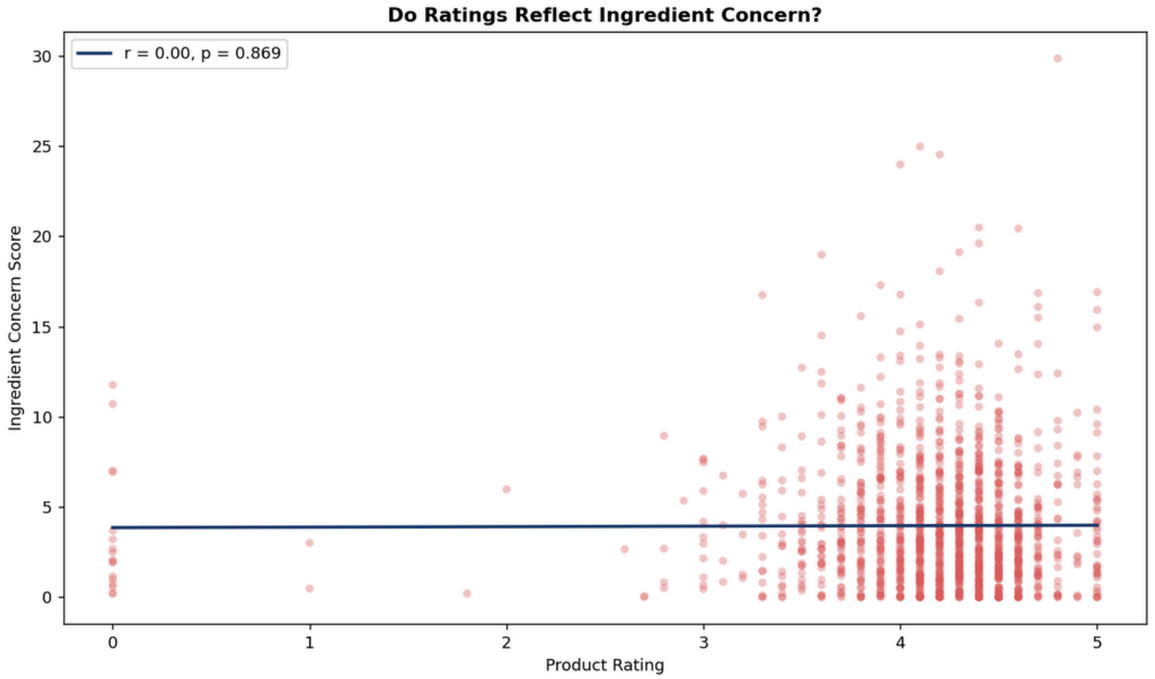
Concern scores are low-to-moderate across all price tiers.

Budget products have the lowest median, but include extreme outliers.

Mid-Range and Premium products show slightly higher median scores and wider spreads.

Premium products do not consistently show lower concern scores than cheaper products.

Finding 2



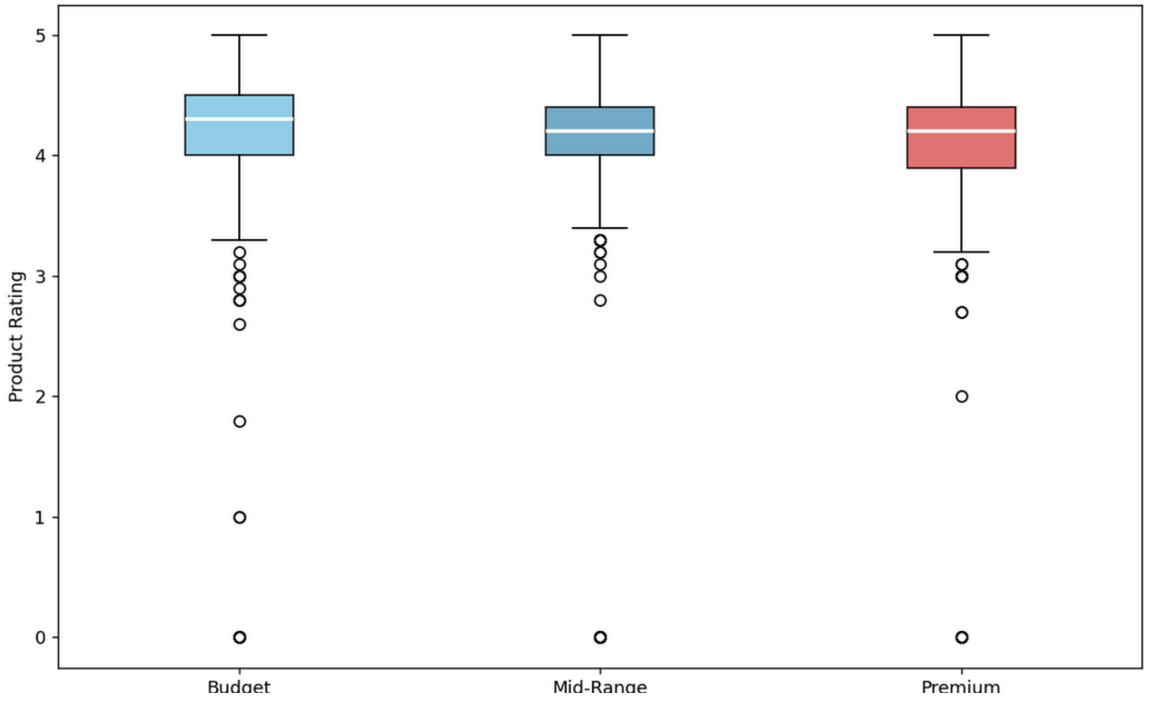
There is no meaningful linear relationship (r=0.00) between rating and concern score.

Most ratings are clustered between 4 and 5, limited data on 0-3 star products.

Concern scores vary widely even among highly rated products.

Implication: Consumer ratings do not appear to reflect ingredient concern. Ratings may capture product experience more than formulation-level concerns.

Finding 3



Ratings are similarly high across all price tiers.

Median ratings are around the low 4's across all tiers.

Budget products have more low-rated outliers compared to other tiers, however most are still rated highly.

Premium products do not receive meaningfully higher ratings than lower-priced products.

Implication: Price does not strongly explain consumer satisfaction in this dataset.

Conclusion of Study

Research Question 1: Do higher-priced products have lower ingredient concern scores?

- **Hypothesis:** Higher-priced products would have lower concern scores.
- **Finding:** Higher-priced products show slightly higher concern scores, but the relationship is weak.
- Higher price does not reliably indicate a cleaner or lower-concern formulation.

Research Question 2: Do higher-priced products receive higher ratings regardless of concern score?

- **Hypothesis:** Higher-priced products would receive higher consumer ratings.
- **Finding:** Ratings remain high across Budget, Mid-Range, and Premium products.
- Ingredient concern score has no meaningful correlation with rating.

Broader Implications

Price-Quality Relationship:

- The price-quality rule of thumb can still be useful, but higher price does not automatically mean a better formulation.
- Lower-priced products may still work well for specific skin concerns and can be valid alternatives to premium products.

Reasons Behind Price Tags:

- Formulation is not always the main reason a product is considered high-end.
- Premium pricing may also reflect brand image, product experience, packaging, stability, texture, or longevity.

Broader Implications

Product Ratings:

- Consumer ratings are helpful, but they are subjective and do not show the full product experience.
- A single rating does not explain why someone gave a product 4 or 5 stars, or whether it worked for their specific skin type.

Reflection & Closing

Thank you for listening!

References

Boyle, P.J. and Lathrop, E.S. (2009), Are consumers' perceptions of price–quality relationships well calibrated?. International Journal of Consumer Studies, 33: 58-63. https://doi.org/10.1111/j.1470-6431.2008.00722.x